I.MDL COURSE INTRODUCTION
INTELLIGENT MACHINE DESIGN LAB

Dr. rer. nat. Mahathi Anand and Dr.-Ing. Lennart Trösken

IMDL coordinators and lecturers

Technical University of Munich

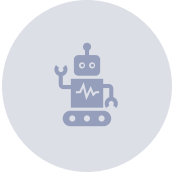
Munich Institute of Robotics and Machine Intelligence (MIRMI)

München, 2026

WHAT IS I.MDL?

- A project-based course where you team up to develop and build a complex, autonomous robotic system that has search and rescue capabilities.

- Teams test and compete in a robotics test arena, challenging each other across key performance metrics



Brought to you by Munich Institute of Robotics and Machine Intelligence (MIRMI), TUM, Unternehmer TUM and robo.innovate



A hot spot for a new generation of innovative engineers and tech-entrepreneurs.



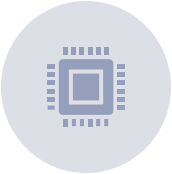
State-of-the art engineering and software design and solutions



Development of powerful robotic systems that can perform complex tasks autonomously



Basic theoretical foundations → Advanced algorithms → Product design



Equipped with all necessary hardware and fabrication facilities

LEARNING OBJECTIVES



Develop multi-disciplinary design skills for mechatronic systems



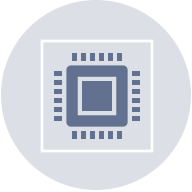
Enable students building own projects from scratch



Study how to develop an innovative idea to a fully-fledged product



Cooperate and communicate in diverse/interdisciplinary teams effectively



Prepare students to tackle contemporary societal, environmental and economical problems



Take your product beyond the scope by identifying relevant market solutions

COURSE SCHEDULE AND STRUCTURE

- Regular Meetings: Every Wednesday
  15/04/2026 - 15/07/2026

- Structure:
  - Self-study
  - Occasional tutorials
  - Online lectures and materials
  - Project tasks



TEACHING TEAM



Dr. rer. nat. Mahathi Anand
Coordinator and Lecturer

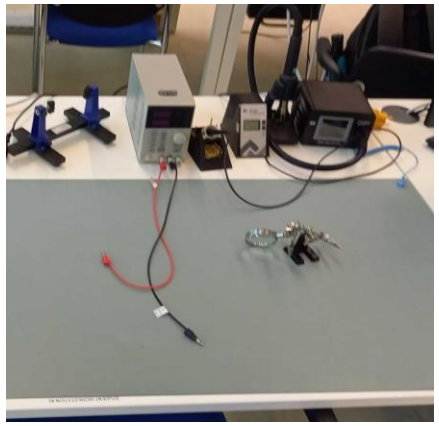


Dr.-Ing. Lennart Trösken
Coordinator and Lecturer

LAB INFRASTRUCTURE

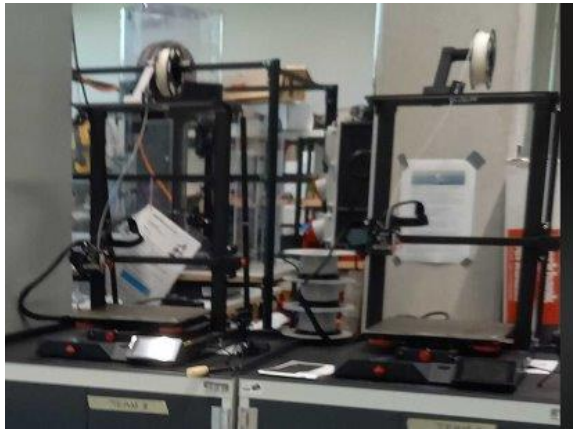
- I.MDL Facilities with Computers
- Electronics Workstation
- Mechanics Workstation
- 3D Printing Workstation
- Mobile robot platform
- Hardware

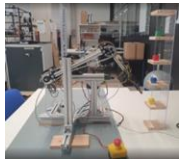








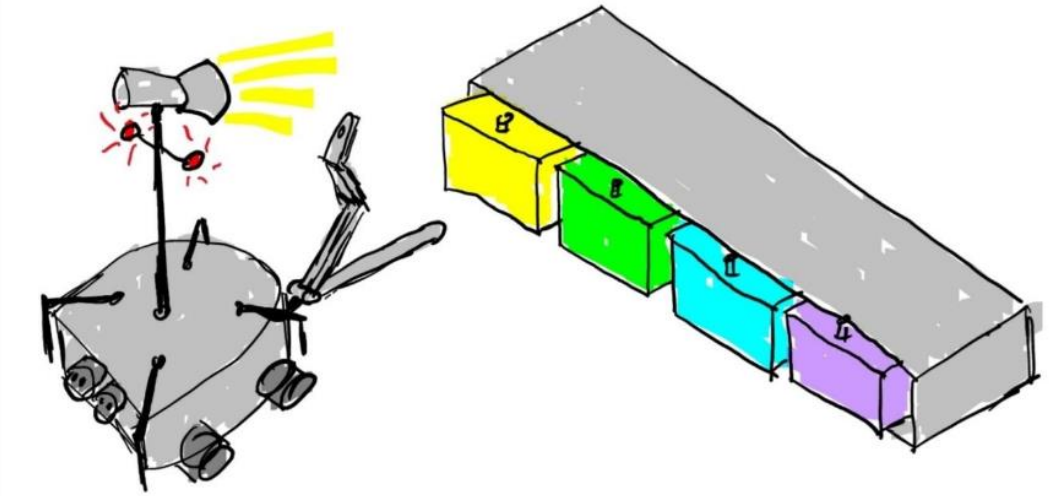








PROJECT DESCRIPTION



- Design a small mobile robot with 'search and rescue capabilities' that can support in mass accidents/catastrophes
- Mobile robot should transport 'casualties' (wooden blocks) from one place to another while maneuvering 'dangerous areas' (obstacles)

HARDWARE MODULES

iRobot Create3 Platform consisting of integrated hardware:

1. ROS2 Compute Unit
2. Cliff sensors, docking sensors, IR sensors
3. Access via Bluetooth/Ethernet/WiFi
4. Optical odometry



HARDWARE MODULES

Master Processing Unit

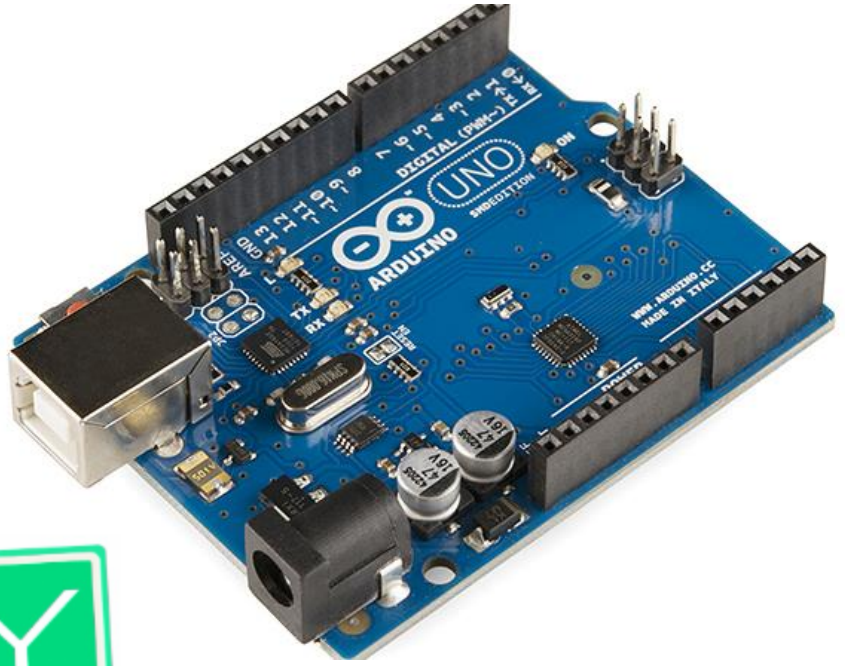
- Jetson Nano Orin single board computer (SBC) for wireless communication to the robot and its sensors
- Provides main computing functionality and communication with the locomotion module (iRobot create 3 platform) and low-level controller (for manipulator) via ROS2



HARDWARE MODULES

Low-Level Controller

- Microcontroller (Arduino Uno) communicates with the SBC via ROS2 and generates the necessary signals to the custom peripherals (manipulator servos, gripper actuators, light beacons, etc.)
- Optional: Use STM32/SAME54 Xplained Pro development board



2 DoF Robot Arm with Gripper

- Robot arm + gripper mechanism and actuation drive. Control signals come from low-level controller. Gripper should lift 4x4 wooden blocks



HARDWARE MODULES

Localization Module

- The module allows to localize the robot within the arena.
- It comprises of IR sensors from icreate3 (IR localization), Intel Realsense camera (Visual SLAM), and/or SLAMTEC C1 Fusion LIDAR (lidar SLAM)





Light Beacon

- allows the robot to provide a light beam (Power LED) into a position it is commanded to. It should be able to redirect the light beam within 180°.

SOFTWARE PLATFORM

Robot Operating System

- ROS2 serves as the primary communication platform between the onboard controller of iRobot create 3, the main processing board, and the low-level controllers.
- It serves as the interface between the custom sensors and actuators as well.
- Use ROS2 Humble with Ubuntu 22.04 for best compatibility with iRobot create3



Embedded C or Arduino IDE

- For low-level controller programming
- Embedded C – microcontroller programming
- Arduino IDE – Arduino UNO

SOFTWARE PLATFORM

Computer Aided Design (CAD)

- For the design of 2 DoF robot arm manipulator with gripper
- Can use any CAD software (e.g. Solidworks)

Simulation Platforms

- Gazebo (with ROS2) can be used for testing planning and control algorithms for the robot
- MATLAB Simulink can be used for the modeling and simulation of manipulator.

PROJECT OBJECTIVES

Mobile Robot

- The iRobot create3 mobile platform should be able to move forward, backward, and rotate
- It should report its status variables (location, velocity, etc.)
- It can detect its absolute position in the test arena
- It can move to a specified location
- It can detect obstacles as well as objects of interest (wooden blocks)



Map the arena via IR localization/LIDAR SLAM/Visual SLAM for localization and navigation



Design an object detection algorithm for determining the colour of the blocks with a light beacon to move to the object and light up for improved color detection.



Develop a path planning algorithm and collision avoidance algorithm to ensure safe navigation



Develop a high-level planning algorithm for task sequencing (navigate --> pick --> navigate --> place --> return)

PROJECT OBJECTIVES

2 DoFs Robot Arm + Gripper

- Arm can move from a given position/orientation to another
- The gripper can orient itself in front of an object
- The gripper can pick objects directly in front of it and can drop it in another place



Design and 3D print a prototype of a 2 DoF robot arm + gripper using CAD software



Develop a trajectory planning algorithm for the robot arm's pick and place task via inverse kinematics



Implement feedback control loops for precise actuation and locomotion of the manipulator

PROJECT OBJECTIVES

Mobile Manipulator

- Robot can report its status variables (Position, gripper status, etc.)
- Robot can move forward, backwards and rotate.
- Robot can detect its absolute position within the arena.
- Robot can move to a specified location.
- Detect pos. of objects directly in front of the gripper.
- Can grasp a box in front of the robot
- Rotate the light beacon to the selected position.
- Turn on/off the light.
- Drop the cube in a selected box (A or B)



Integrate the mobile platform with the robot arm

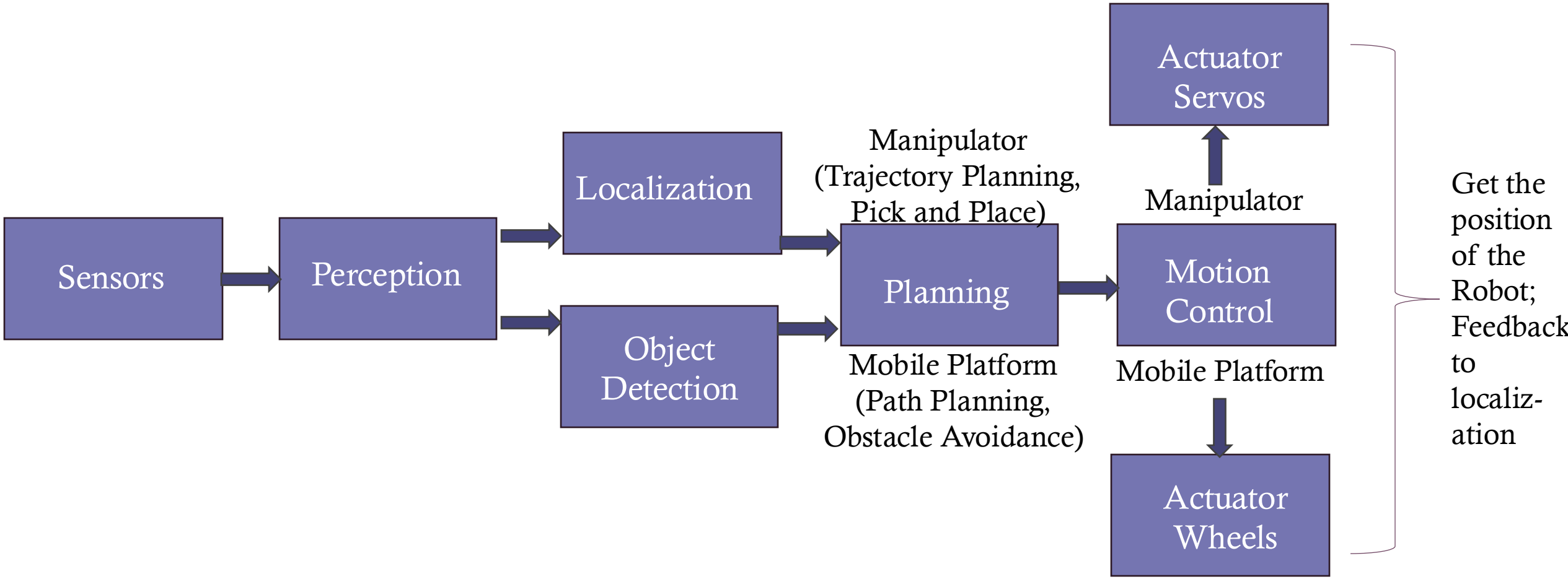


Develop path planning for navigation of mobile robot with pick and place of the manipulator

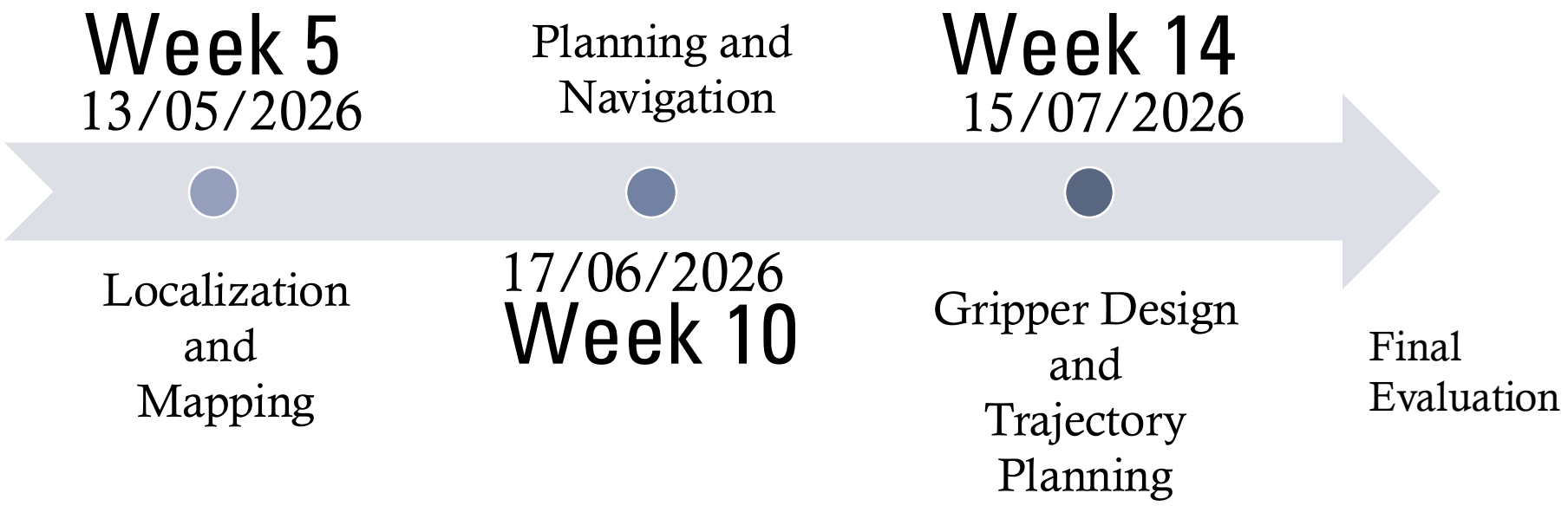


Ensure that the system is fully-functional and can perform perception, object detection, task sequencing, and pick and place while maintaining obstacle avoidance

PRELIMINARY SYSTEM DESIGN



MILESTONES



On Milestone Day:

- 5 minute presentation/report on progress
- Discussion of innovative ideas applied to the project
- Simulations and practical demonstrations on the progress

TASKS (MILESTONE 1)

Hardware Tasks

- Explore hardware modules
- 3D print sensor mounts and ensure sensor calibration
- Maintain power and data connections, proper communication between mobile platform, SBC, and sensors
- Integrate all software modules and ensure proper mapping in real-time
- Make the robot move and localize.

Software Tasks

- Explore ROS2, ROS2 gazebo
- Explore ROS2 APIs, find relevant information on motion control of mobile platform
- Design full system architecture and plan for all modules
- Develop SLAM algorithm enabling robot to map the arena
- Simulation of the SLAM algorithm via simulation

Management Tasks

- Formation of teams and distribution of tasks based on expertise
- Set project timeline, and manage tasks and deliverables
- Coordinate resources and integrate between members
- Track weekly progress and update timelines when necessary

TASKS (MILESTONE 2)

Hardware Tasks

- Verify proper sensor calibration and accuracy
- Verify operation of the localization module
- Log any errors and debug them in a modular fashion
- Integrate planning module with localization module for improved planning + perception
- Integrate light beacon for improved object detection

Software Tasks

- Explore and implement object detection algorithms including obstacles and color of the objects
- Explore and implement navigation algorithms with perception
- Develop a task sequence algorithm to execute navigation as a sequence: Position A -> Position B -> Position C -> Position A
- Implement the algorithms via ROS2
- Test on simulation

Management Tasks

- Manage tasks and deliverables
- Track milestones and integrate lessons from previous phases
- Plan for next phases and intregrate results early
- Coordinate resources and integrate between members
- Track weekly progress and update timelines when necessary

TASKS (MILESTONE 3)

Hardware Tasks

- Design 2 DoF robot arm and gripper by considering joint angles, link lengths, gripper size, and torque requirements
- 3D print, assemble, and mount the manipulator on mobile base
- Test actuator functionality and ensure proper operation with the motion control module

Software Tasks

- Design 2 DoF robot arm using CAD software
- Develop trajectory planning for manipulator
- Implement commands for the actuators using ROS2 and low-level controller
- Simulate manipulator behavior in MATLAB/ROS2
- Ensure reliable operation through proper logging and debugging

Management Tasks

- Manage tasks and deliverables
- Track milestones and integrate lessons from previous phases
- Plan for next phases and intregrate results early
- Coordinate resources and integrate between members
- Track weekly progress and update timelines when necessary

TASKS (FINAL DEMONSTRATION)

Hardware Tasks

- Fully integrate the manipulator with the mobile platform
- Refine the prototype design and make any necessary adjustments
- Validate the system and perform necessary tests in various environment scenarios
- Validate system behavior, log any errors and debug in a modular way

Software Tasks

- Integrate all modules – perception, navigation, manipulation
- Combine mobile platform motion control with manipulator control
- Develop a unified task sequencing framework with manipulator to perform
  (navigate -> pick -> navigate -> place -> return)
- Repair, finetune and validate all modules for correct operation
- Evaluate performance

Management Tasks

- Manage tasks and deliverables
- Consolidate and report all technical work
- Prepare detailed technical documentation of the project prototype
- Plan integration milestones
- Ensure system reliability
- Prepare final presentation with tasks, achieved deliverables and reports

OPTIONAL TASKS

- Multi-sensor Fusion: Use two or more sensors and combine them for perception and planning
- Path and Trajectory Optimization: For path and/or trajectory optimization, plan for minimum time and/or minimum power consumption

- Optional tasks are not specifically graded or evaluated, but demonstration of these tasks will give the teams "bonus"

IMPORTANT CONSIDERATIONS

- **Define clear roles and responsibilities**: Assign ownership of specific subsystems or tasks (e.g. perception, planning, control, hardware design, documentation and reporting) based on specific expertise and previous knowledge
- S**et measurable deliverables**: Break big goals into smaller, manageable, and verifiable outcomes
- **Integrate early and often**: Don't wait until the end for hardware integration, software or even ROS2 nodes
- **Test incrementally**: Each module should be validated at least once as standalone and once before integration. Similarly submodules should all be validated before final module validation.
- **Time for Debugging:** Reserve time for integration testing, tuning, and debugging
- **Version contro**l: Use Git and commit frequently with meaningful, readable and trackable information
- **Regular check in and progress tracking**: Meet between team members often (daily/weekly) to track and manage outcomes

PROJECT TRACKING CHART

Sr. No	Deliverable	Description	Owner	Status
1.	Project pipeline			
2.	System architecture			
3.	Planning Module			
4.	…			

EVALUATION AND GRADING

- The grade of each team member will be calculated based on different factors
  - System-based (for each team member identical)
    - (20%) Design & Production quality
    - (20%) Basic functionality
    - (20%) In-Field challenge
  - Individual
    - (15%) Milestone meetings (5% each milestone)
    - (15%) Final presentation (5% presentation and visualization, 5% technical details, 5% design and algorithm reasoning)
    - (10%) Self-Evaluation
- Milestone Meetings: 5 min presentation/simulation, project deliverables report
- Final Presentation: 10 min presentation on the concept and details of your system
- Optional Tasks: Extra 5% per task up to 95%

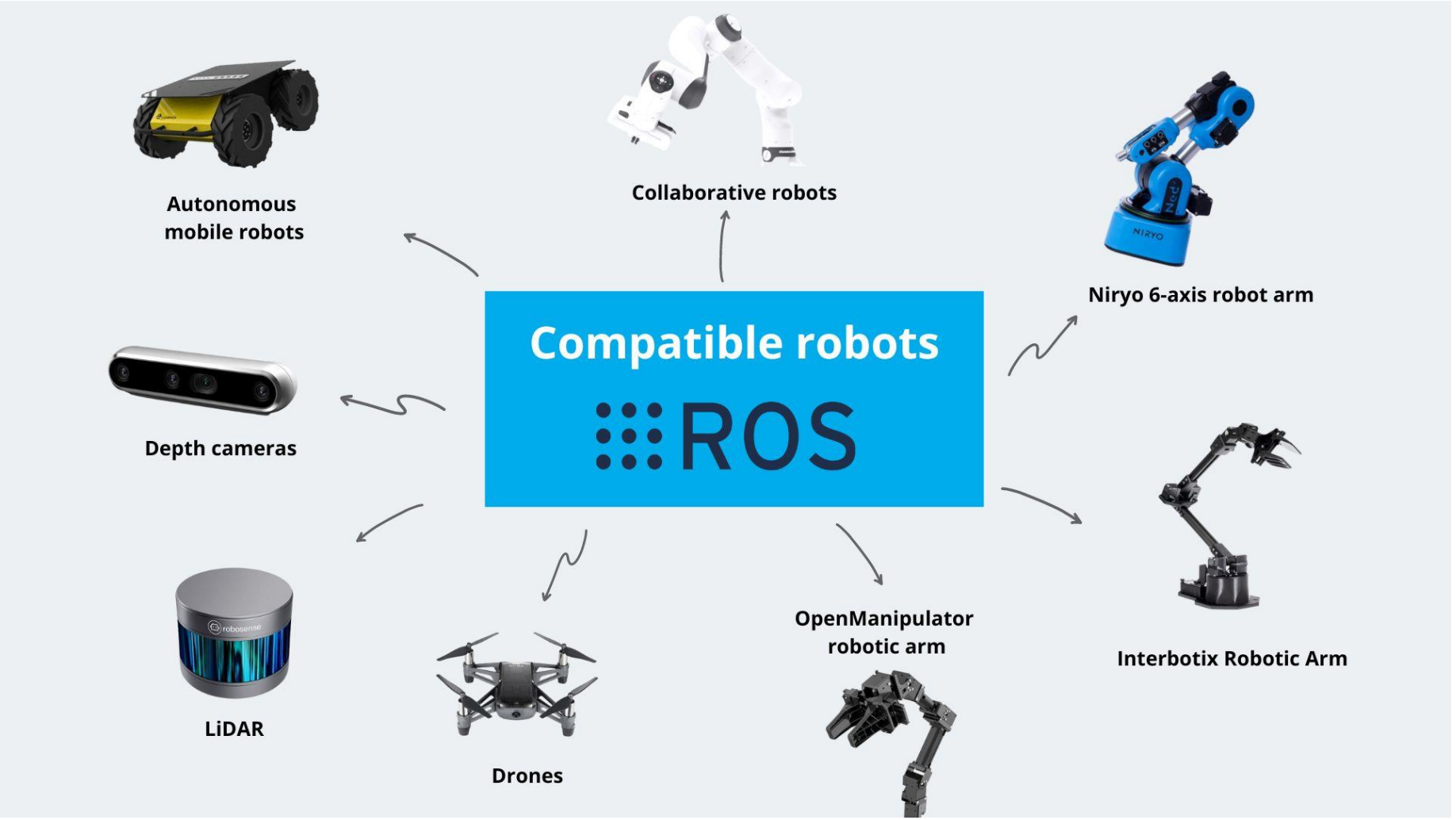
TUTORIALS AND SUPPORT

- Tutorials on Path Planning/SLAM (Next week)
- Tutorials on Manipulator Design/Motion Control (Next week)
- Possibly more tutorials on other topics relevant to the course (TBD in future)
- Online resources consolidated in **Wiki** (I.MDL Student Wiki) including online lectures/tutorials/data sheets and manuals
- Please use self-study mechanism as much as possible to solve your issues
- For any additional technical help from other team members and lecturers – Use forum in **TUM Moodle**
- **Slack Channels** for team-wise discussions, and general channel for organizational announcements

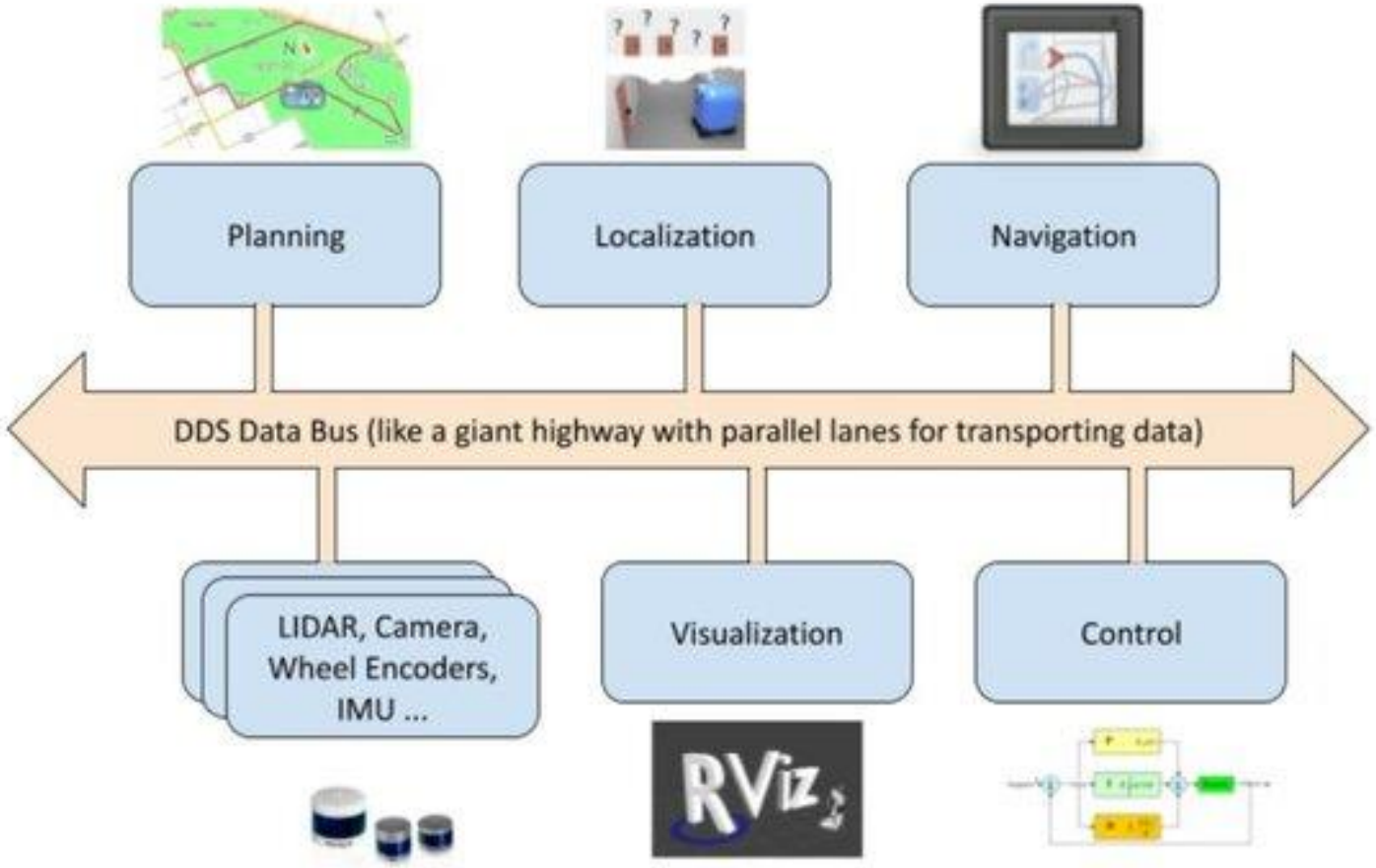
FIRST STEPS

- **Form yourself into teams (3-4 team, 4-5 students per team) --> moodle, link to a spreadsheet**
- Set up ROS2 - Humble on Ubuntu 22.04 for best compatibility with iRobot create3 mobile platform and NVIDIA Jetson Orin Nano
- Install an IDE for coding, e.g. Visual Studio Code (VS Code)
- Learn about ROS2 nodes, topics, actions, servers and run your first ROS2 package
- Play around with iRobot Create3 Platform – Update firmware, run example codes, and get the robot moving!
- Look into LIDAR datasheet documentation, print the mounts and start looking into LIDAR SLAM algorithms before the next tutorial

ROS2


Autonomous mobile robots
Collaborative robots
Niryo 6-axis robot arm
Compatible robots
ROS
Depth cameras
LiDAR
Drones
OpenManipulator robotic arm
Interbotix Robotic Arm

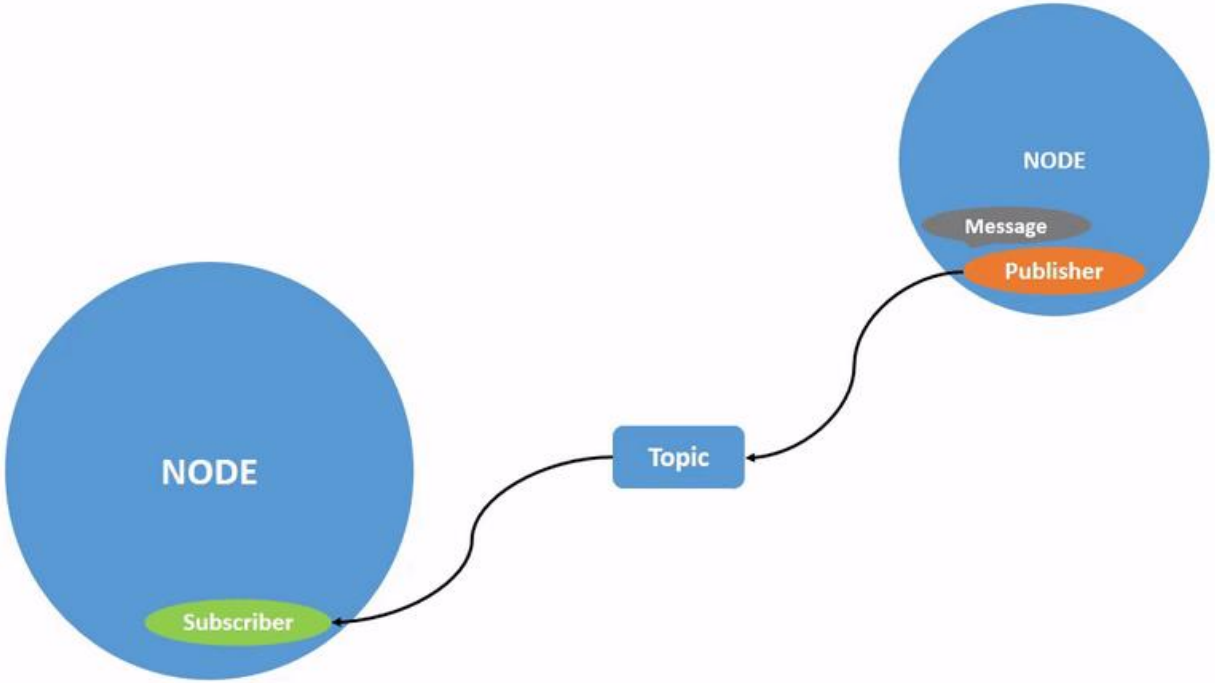
ROS2



- We use ROS as a tool to let multiple programs communicate with each other
- Instead of having one huge and complex program, we can write several small programs that can talk to each other
- Distributed: can be run on multiple computers, communicate over network
- De facto standard for robot programming

ROS2 - SERVICES

ROS 2 breaks complex systems down into many modular nodes. Topics are a vital element of the ROS graph that act as a bus for nodes to exchange messages.

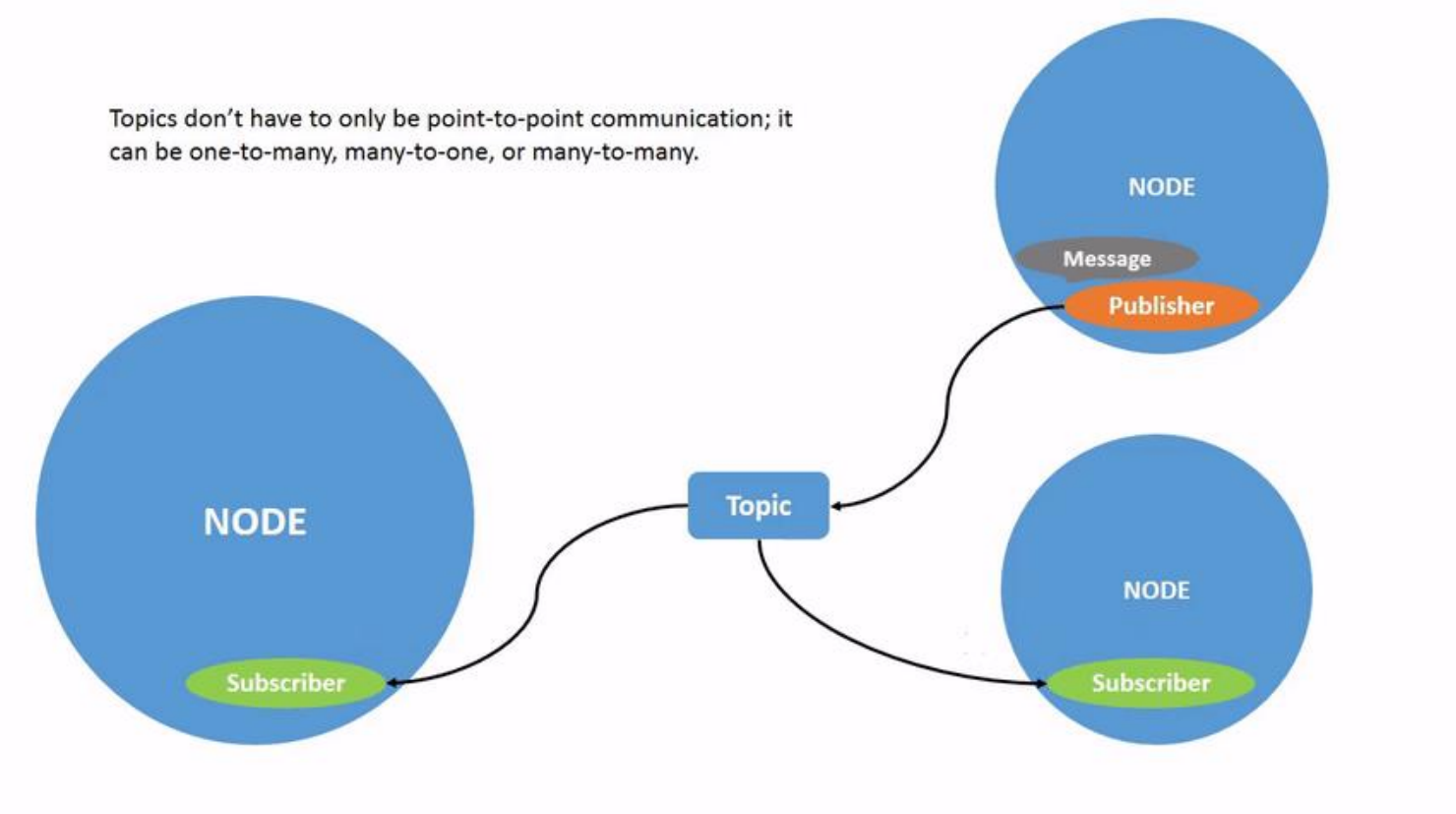



```
ros2 topic pub /chatter std_msgs/msg/String "data: My name is …."

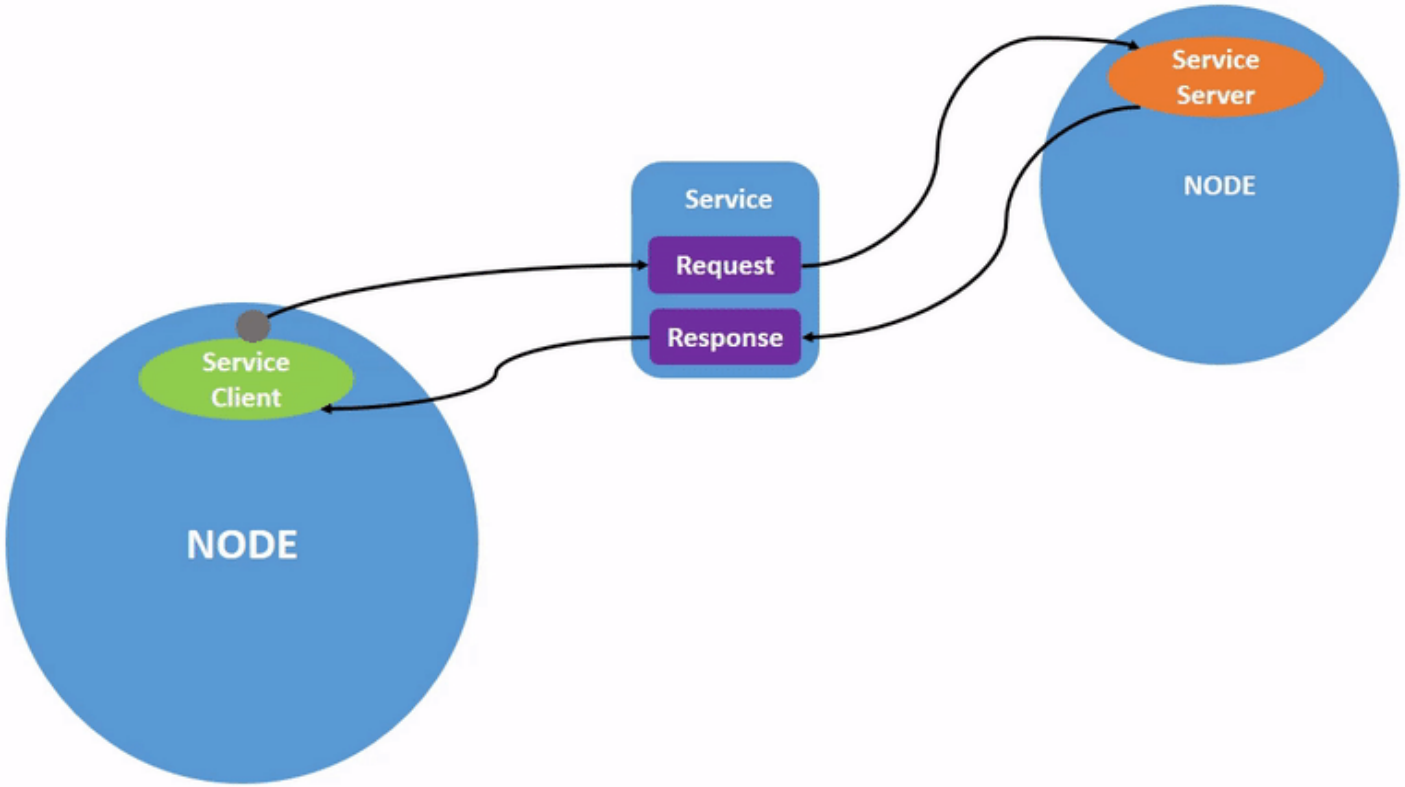
ros2 topic echo /chatter
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ROS2 - SERVICES

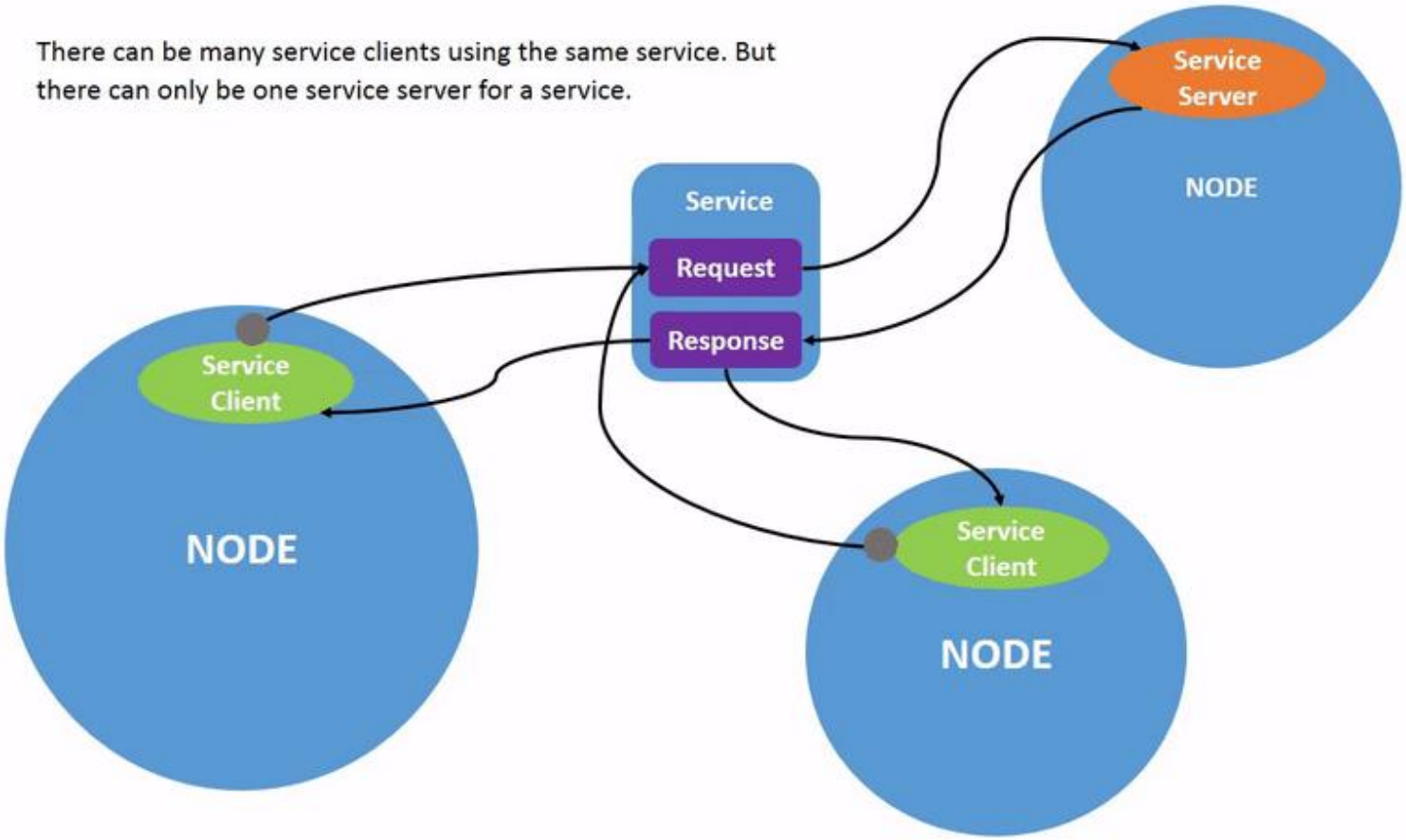
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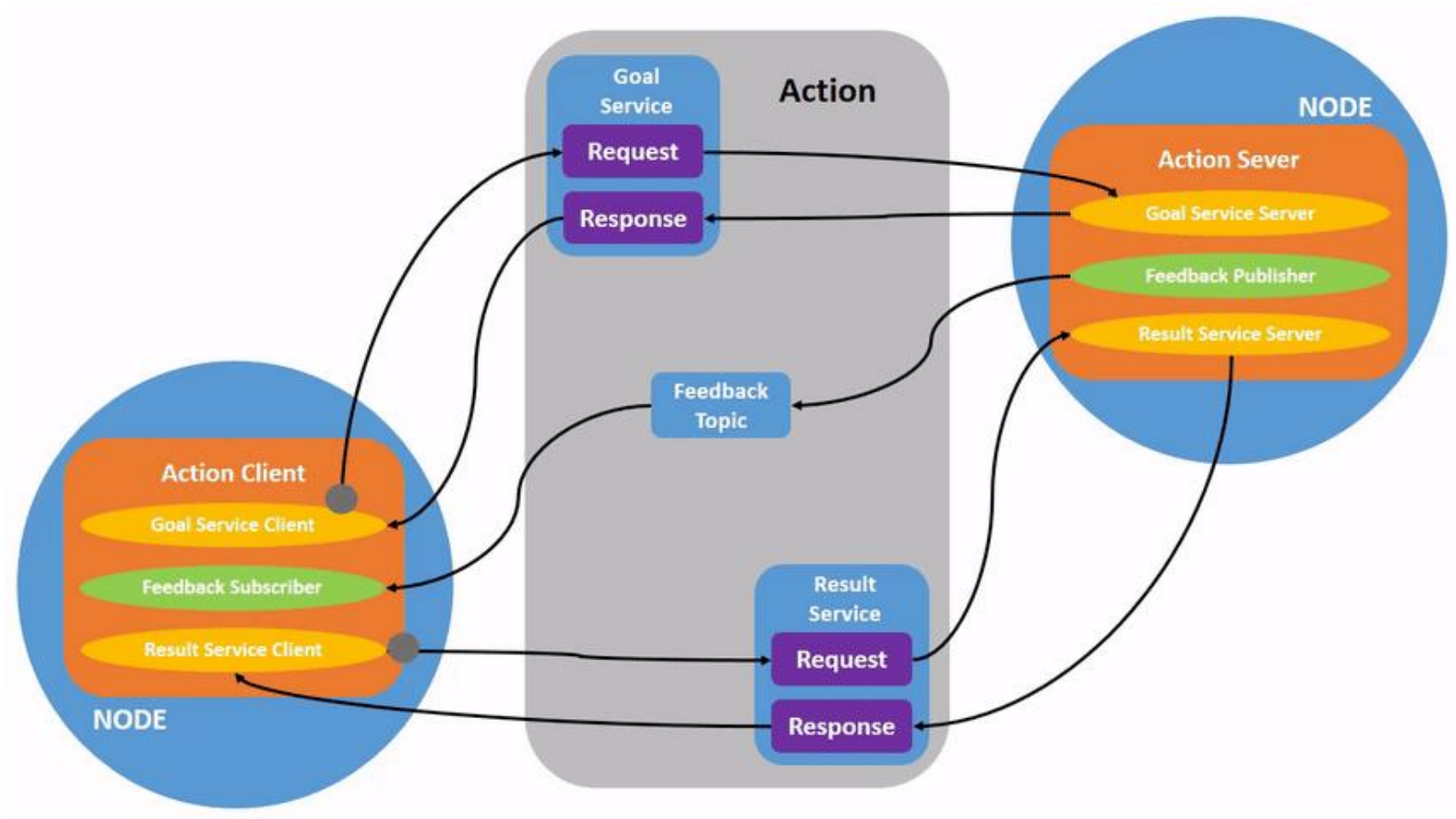
ROS2 - SERVICES



ROS2 - SERVICES



ROS2 - ACTIONS



THANK YOU